

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$f(x) = (x + 6)(x - 4)$

If the given function f is graphed in the xy -plane, where $y = f(x)$, what is the x -coordinate of an x -intercept of the graph?

Correct Answer: -6, 4

Rationale

The correct answer is either -6 or 4 . The x -intercepts of a graph in the xy -plane are the points where $y = 0$. Thus, for an x -intercept of the graph of $y = f(x)$, $0 = f(x)$. Substituting 0 for $f(x)$ in the equation $f(x) = (x + 6)(x - 4)$ yields $0 = (x + 6)(x - 4)$. By the zero product property, $x + 6 = 0$ and $x - 4 = 0$. Subtracting 6 from both sides of the equation $x + 6 = 0$ yields $x = -6$. Adding 4 to both sides of the equation $x - 4 = 0$ yields $x = 4$. Therefore, the x -coordinates of the x -intercepts of the graph of $y = f(x)$ are -6 and 4 . Note that -6 and 4 are examples of ways to enter a correct answer.